WashU McKelvey Engineering

Spring 2026 MEMS 3430 Design of Thermal Systems

Design Homework 1 - Brayton Topping Cycle - Submission #2

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Submission Date: Monday, February 9, 2026

Assumptions and boundary conditions

Assumptions.

- Air is assumed to behave as an ideal gas. It is inputted and exhausted at ambient air temperature/ pressure.
- The cycle is assumed to be perfectly isentropic, where any inefficiencies are accounted for within the turbine and compressor isentropic efficiencies. C_p can be treated as a constant value for this assumption.
- There are no heat gains/losses inside the system as the gas goes from one state to another internal to the system. Heat is only added or removed at the high and low temperature heat exchangers.
- Pressure is constant across the high and low temperature heat exchanger, neglecting any major or minor losses.
- Heat at the heat exchangers is perfectly efficient. I.e. there is exactly 30MW of heat removed from the low temperature heat exchanger and input into the bottoming cycle.
- There are no mechanical losses in the compressor or the turbine
- The compression ratios of the system are realistically between 5 and 30.
- The pressure differential across the compressor and turbine is equal to the compression ratio
- The second half of the assignment looks at the solar power tower, the DNI pulled from a literature search is said to be independent on day to day conditions (as it is a daily average) and independent of incidence angle and shows perfect solar to thermal effects. This will give us an idealized maximum value for the system.

Boundary Conditions.

- An important boundary condition is the heat leaving this cycle between state four and state five, this heat removal allows the energy balance to be complete. This heat is 30MW.
- Air enters the compressor at 300K
- Air enters the turbine at 1200K
- Air leaves the low temperature heat exchanger at 500K
- A realistic set of compressor and turbine efficiencies was pulled from a NASA paper that went into details of modeling a gas turbine system. Those values are:

  $[\eta_c, \eta_t] = [0.87, 0.85]$ [1]

- The second half of this assignment for the solar power tower approach requires selecting a location and determining the solar irradiance. The location I am selecting is:

  Tuscon, Arizona

  Where the average monthly Direct Normal Irradiance (DNI) is 7.62 kWh/m2/day (or 0.3175kW/m2). [2]

T-s & P-v diagrams of Brayton Cycle

T-s Diagram of Brayton Cycle. Figure 1 is the Temperature - Entropy diagram for the Brayton topping cycle. Air enters the cycle at 300K, then enters the turbine at 1200K, then exits the system at 500K. The compressor and turbine are originally assumed to be perfectly isentropic processes, before the efficiency is applied.

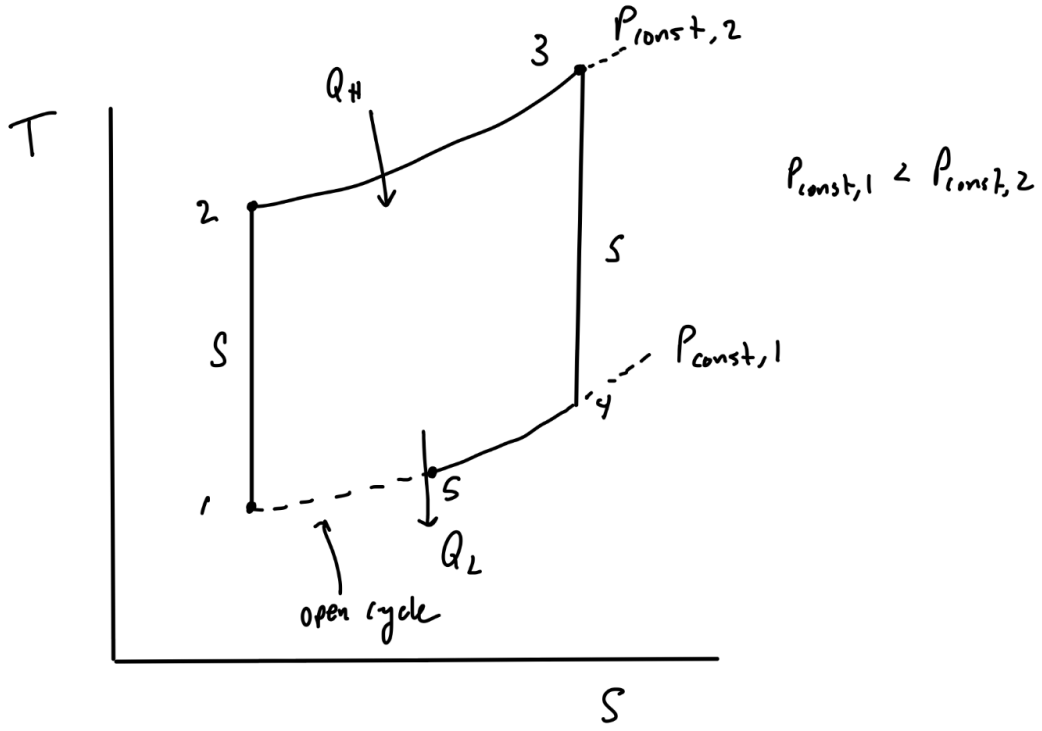


Figure 1 Temperature vs Entropy Diagram for the Brayton Topping Cycle

0.1 P-v Diagram of Brayton Cycle. Figure 2 is the Pressure - Volume diagram for the Brayton topping cycle. The pressure is entirely dependent on the compression ratio, which is optimized for in this design homework. Similarly, the pressure is constant across the high and low temp heat exchangers.

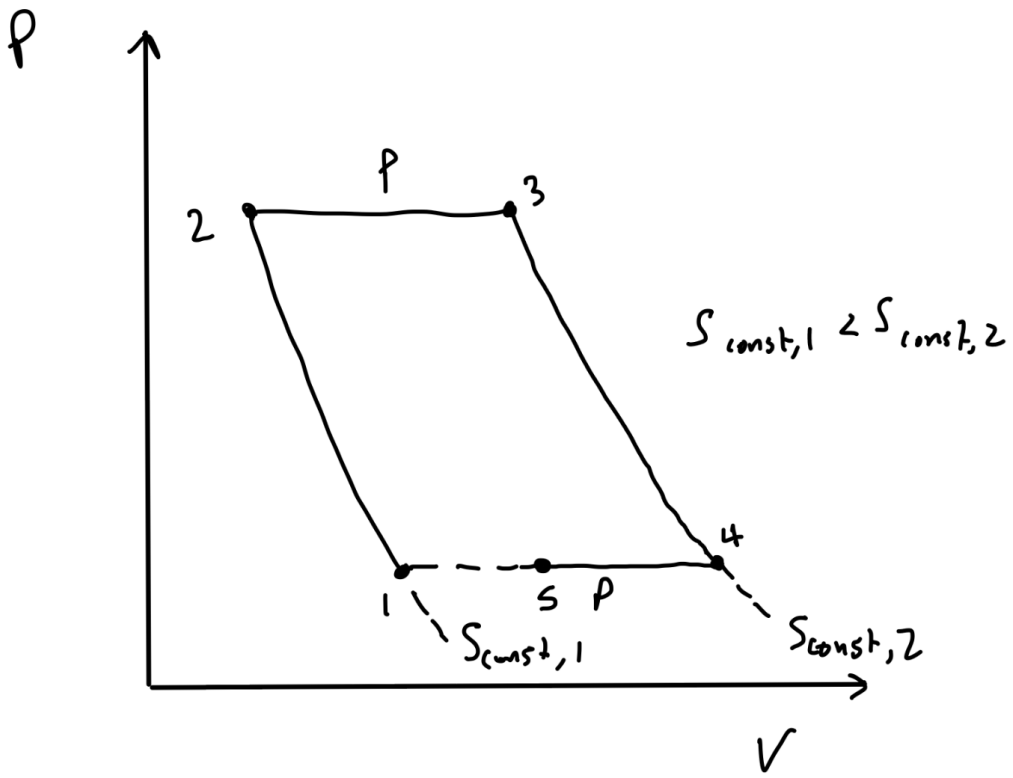


Figure 2 Pressure vs Volume Diagram for the Brayton Topping Cycle

Main Equations Used

Turbine. The ideal temperature at the exit of the turbine, can be found using Eq. 1. Where T_3 is the turbine inlet temperature, 1200K (given in problem statement), r_p is the compression ratio, and k is the isentropic coefficent.

$$T_{4,ideal} = T_3(\frac{1}{r_p})^{\frac{k-1}{k}} \tag{1}$$

Next, the real temperature T_4 can be found using Eq. 2. Where η_t is the efficiency of the turbine, laid out in the boundary conditions section.

$$T_4 = T_3 - \eta_t(T_3 - T_{4,ideal}) \tag{2}$$

The last equation for the turbine is the specific work done using Eq. 3. Where C_P is the specific heat and the temperatures are previously defined.

$$W_t = C_p(T_3 - T_4) \tag{3}$$

Compressor. The ideal temperature at the exit of the compressor, can be found using Eq. 4. Where T_1 is the compressor inlet temperature, 300K (given in problem statement), r_p is the compression ratio, and k is the isentropic coefficent.

$$T_{2,ideal} = T_1(r_p)^{\frac{k-1}{k}} \tag{4}$$

Next, the real temperature T_2 can be found using Eq. 5. Where η_t is the efficiency of the compressor, laid out in the boundary conditions section.

$$T_2 = T_1 - \frac{1}{\eta_c}(T_1 - T_{2,ideal}) \tag{5}$$

The last equation for the compressor is the specific work done using Eq. 6. Where C_P is the specific heat and the temperatures are previously defined.

$$W_c = C_p(T_1 - T_2) \tag{6}$$

High Temp Heat Exchanger. The specific heat into the heating portion of the cycle can be calculated using Eq. 7.

$$q_{in} = C_p(T_3 - T_2) \tag{7}$$

Low Temp Heat Exchanger. The specific heat out into the bottoming cycle can be calculated using Eq. 8. Where C_P is the specific heat, and T_5 is given in the problem as 500K.

$$q_{out} = C_p(T_4 - T_5) \tag{8}$$

Using this q_{out} and the heat rejection boundary condition provided in the problem (30MW), we can solve for $\dot{m}$ using Eq. 9.

$$\dot{m} = \frac{\dot{Q}_{out}}{q_{out}} \tag{9}$$

Net Work by System. Using all these components so far, we can solve for the total net work of the system using Eq. 10.

$$\dot{W}_{net} = \dot{m}(W_t - W_c) \tag{10}$$

Thermodynamic Efficiency of Cycle. Using the high temp heat input and the low temp heat output, we can find the cycle thermodynamic efficiency using Eq. 11.

$$\eta_{th} = 1 - \frac{q_{out}}{q_{in}} \tag{11}$$

Heliosat Calculations. To find the number of 10m x 10m heliostats required, a few calculations are needed first. Firstly, the total required heat input for the high temperature heat exchanger $\dot{Q}_H$ this is found using Eq. 12.

$$\dot{Q}_H = \dot{m} q_{in} \tag{12}$$

To find the amount of total energy from the sun on each heliostat, we can use Eq. 13.

$$\dot{Q}_{heliosat} = DNI \times A_{heliosat} \tag{13}$$

Using these relations, the total number of 10m x 10m heliostats can be calculated using Eq. **??**

$$n = \frac{\dot{Q}_H}{\dot{Q}_{heliosat}} \tag{14}$$

These equations are all utilized to show full walkthrough of the math in the next section.

Calculations

This section is a set of calculations used for the heliosat section of the report where specific numbers are calculated for number of heliosats required and other aspects in the system. The selected compression ratio from the previous section is 19.

$T_{4,ideal} = 1200K(\frac{1}{19})^{\frac{0.4}{1.4}}$	Solving for ideal T_4 using Eq. 1
$= 517.4K$	
$T_4 = 1200K - 0.85(1200K - 517.4K)$	Solving for real T_4 using Eq. 2
$= 619.8K$	
$W_t = 1004.5(1200 - 619.8)$	Solving for specific turbine work using Eq. 3
$= 582.8kW/kg$	
$T_{2,ideal} = 300K(19)^{\frac{0.4}{1.4}}$	Solving for ideal T_2 using Eq. 4
$= 695.8K$	
$T_2 = 300K - \frac{1}{0.87}(300K - 695.8K)$	Solving for real T_2 using Eq. 5
$= 754.9K$	
$W_c = 1004.5(300 - 754.9)$	Solving for specific compressor work using Eq. 6
$= -457kW/kg$	The compressor requires work input, thus it is negative
$q_{in} = 1004.5(1200K - 754.9K)$	The specific heat input required can be found using Eq. 7
$= 447kJ/kg$	
$q_{out} = 1004.5(619.8K - 500K)$	The specific heat output can be found using Eq. 8
$= 120.3kJ/kg$	
$\dot{m} = \frac{30000}{120.3}$	The mass flow can be found using Eq. 9
$= 249.3kg/s$	
$\dot{W}_{net} = 249.3(582.8 - 457)$	The net power of the cycle can be found using Eq. 10
$= 31.4MW$	
$\dot{Q}_H = 249.3(447)$	Find the total heat input required using Eq. 12
$= 111.4MW$	
$\dot{Q}_{heliosat} = 0.3175(100)$	Heliostat heat gen using Eq. 13
$= 31.75kW$	
$n = \frac{111435}{31.75}$	Number of heliostats to satify heat requirement using Eq. 14
$= 3510$ heliosats	

Results & Discussion

Thermodynamic Efficiency vs Compression Ratio. Below in Figure 3 is a plot of the thermodynamic efficiencies as a function of compression ratio for the seven predefined compressor and turbine efficiency combinations.

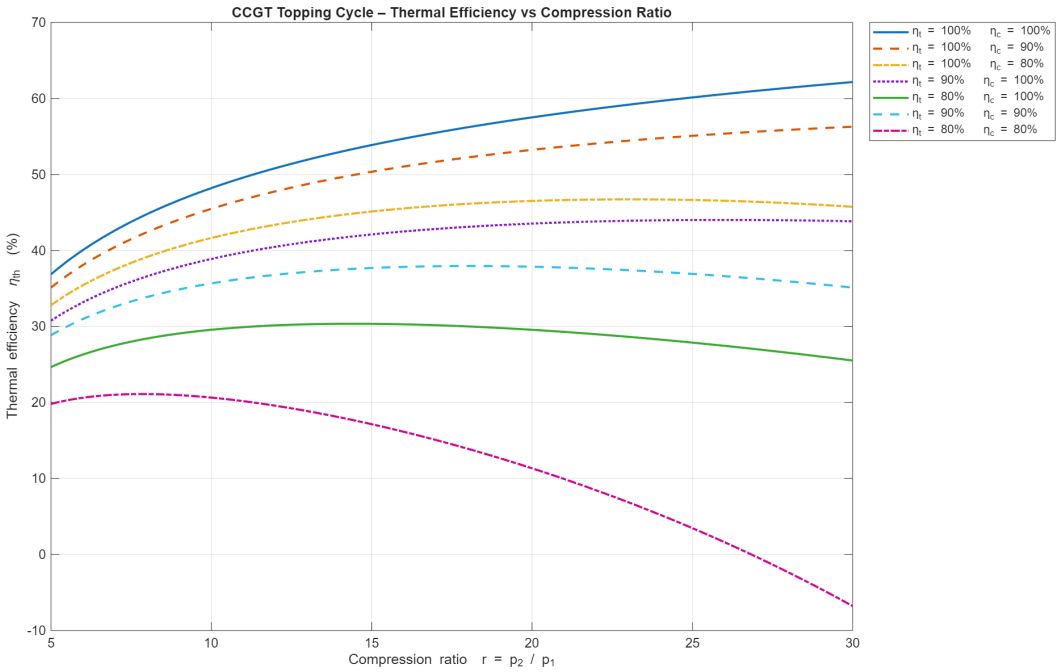


Figure 3 Thermodynamic Efficiency vs Compression Ratio for the Brayton Topping Cycle

The compression ratios vary between 5 and 30 as given in the problem statement. The highest thermodynamic efficiency is when both the compressor and turbine have 100% efficiency, which makes sense because this is the most idealized system. The lowest efficiency when both the compressor and turbine have 80% efficiency, which also makes sense because it is the lowest combination of efficiencies of the set. The turbine efficiency has a higher impact on the thermodynamic efficiency than the compressor efficiency does. The thermodynamic efficiency for when both the compressor and turbine have 80% efficiency goes negative when the compression ratios are high, this is unrealistic for a real useful cycle, but can be explained by a negative net work. If the net work is negative, it takes more work to power the compressor than you are getting out of the turbine, there is a negative efficiency (a non-realistic cycle).

Net Power Output vs Compression Ratio. Below in Figure 4 is a plot of the net power generated as a function of compression ratio for the seven predefined compressor and turbine efficiency combinations.

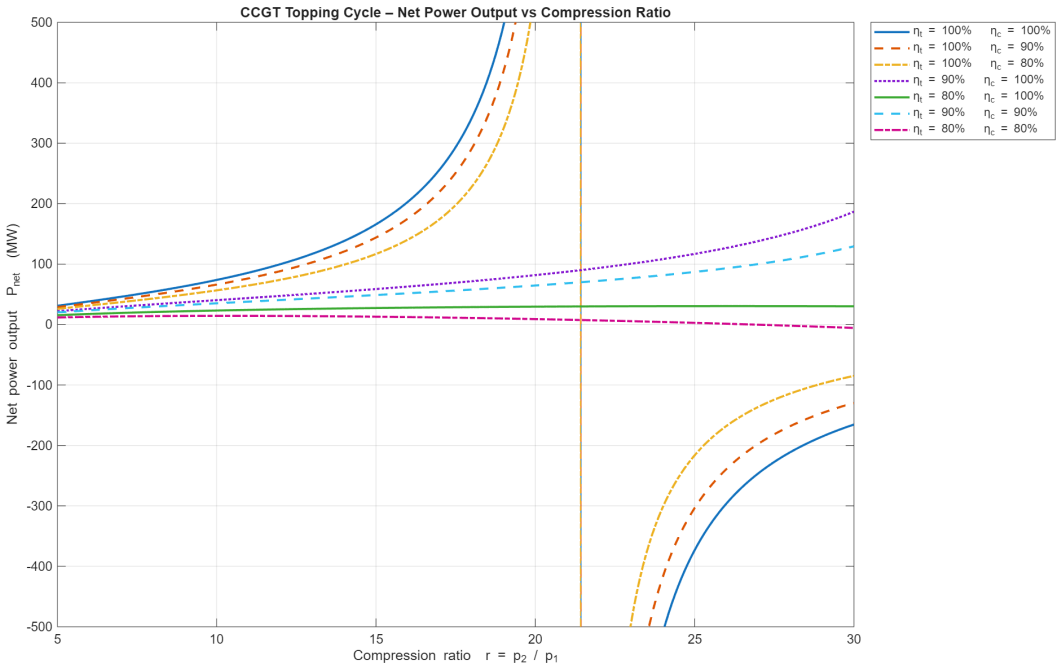


Figure 4 Net Power Output vs Compression Ratio for the Brayton Topping Cycle

At a compression ratio of $\sim$ 21.36, the three efficiency ratios with a turbine efficiency of 100% increase in power output rapidly, then abruptly flip negative and continue with negative net power output. The rapid increase in power output is due to the idealized turbine, showing the significance of an efficient turbine. The rapid increase in power output only appears in the 100% efficient turbine combinations. The net negative power output in itself isn't impossible (more compressor work than turbine work), but upon looking into the individual numbers it is more clear what is happening at these higher compression ratios. As the compression ratio increases for these high efficiency cycles, the temperature T_4 continually drops, until the compression ratio is equal to $\sim$ 21.36 at which point $T_4 < T_5$. This is important, because when $T_4 < T_5$, the mass flow rate calculation becomes negative which in turn makes the net power output negative. This negative mass flow rate is not physically possible for this system.

Optimal Compression Ratio for each Efficiency Ratio. Below in Table 1 is the

Turbine Efficiency (η_t)	Compressor Efficiency (η_c)	Optimal Compression Ratio (r_p)	Thermodynamic Efficiency (η_{th})	Net Power Output (MW)
100	100	21.36	58.3	24868.5
100	90	21.36	53.8	20730.7
100	80	21.36	46.72	15558.5
90	100	26	44.00	126.6
90	90	18.1	37.95	58.1
80	100	14.4	30.36	27.07
80	80	7.8	21.11	13.9

Table 1 Caption

The table above maximized the thermodynamic efficiencies to determine what the optimal compression ratio was. However, in the data where the net work suddenly went negative, due to the negative mass flow this was deemed an impossible cycle and 21.36 was the compression ratio with maximum efficiency. All data suggests that optimizing the turbine should be the engineers number one priority because the net power output and efficiency were higher than their less efficient turbine counterparts.

Thermodynamic Efficiency using real Turbine/Compressor Efficiencies. Below in Figure 5 is a plot of the thermodynamic efficiencies vs compression ratio using the literature values for the compressor and turbine efficiency.

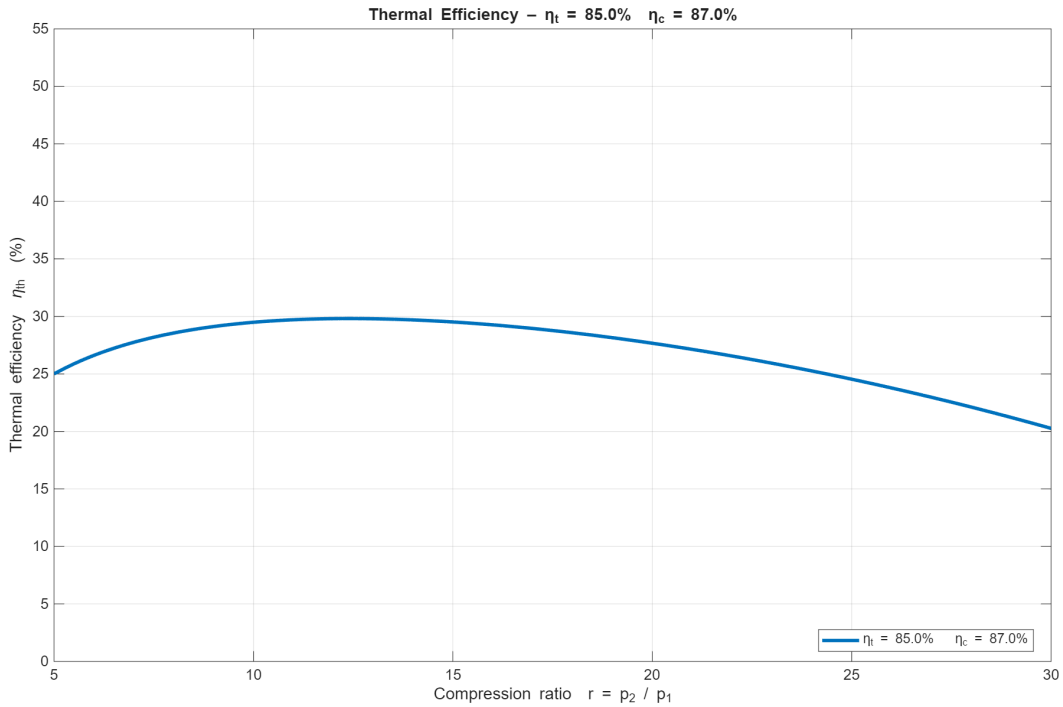


Figure 5 Thermodynamic Efficiency using real Turbine/Compressor Efficiencies for the Brayton Topping Cycle

In the figure above, the thermodynamic efficiency vs compression ratio is plotted using the literature values of compressor and turbine efficiency. The thermodynamic efficiency peaks at a compression ratio of $\sim$ 12.7 at an efficiency of $\sim$ 30%. The thermodynamic efficiency drops to a minimum of $\sim$ 20% at a compression ratio of 30.

Net Power using real Turbine/Compressor Efficiencies. Below in Figure 6 is a plot of the net power output vs compression ratio using the literature values for the compressor and turbine efficiency.

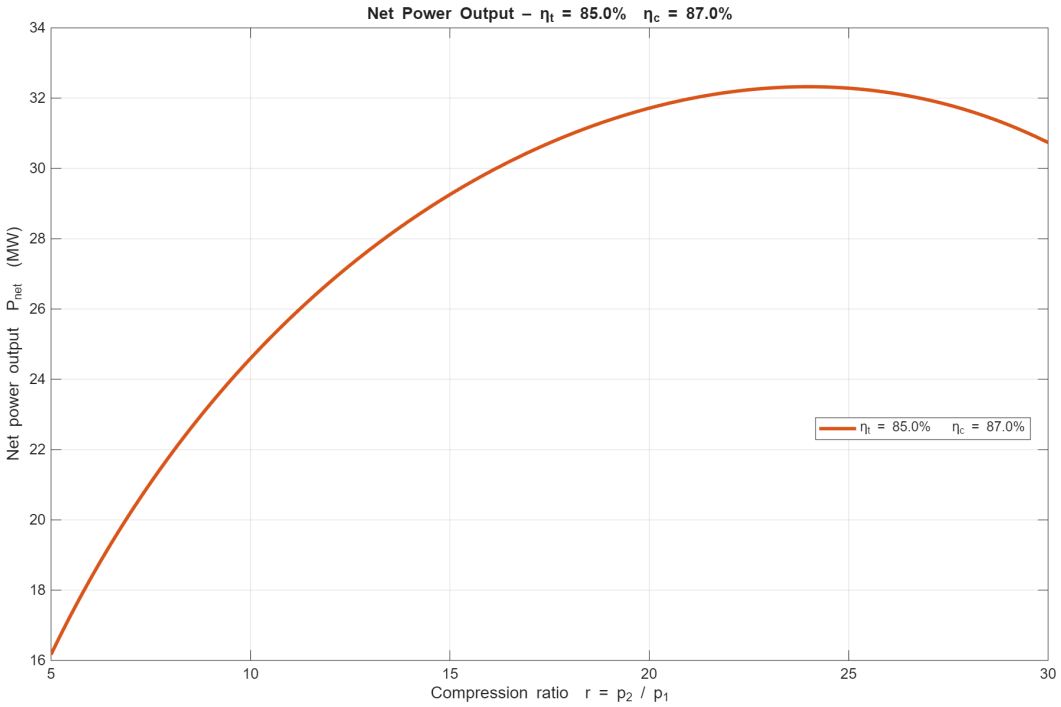


Figure 6 Net Power using real Turbine/Compressor Efficiencies for the Brayton Topping Cycle

In the figure above, the net power output vs compression ratio is plotted using the literature values of compressor and turbine efficiency. The net power output peaks at about $\sim$ 32.3MW at a compression ratio of $\sim$ 23.9. The net power output is at a minimum of $\sim$ 16.2MW at the initial compression ratio of 5, with it flattening out after the peak to $\sim$ 30.1MW at a compression ratio of 30. It is critical to balance both the thermodynamic efficiency and the net power output, which will be done next.

Selection of Compression Ratio. The ranges of ideal compression ratios is initially narrowed down to the ”acceptance” range of 12.7 to 23.9. This range captures both the peak of the thermodynamic efficiency and the peak of the net power output. To balance both the net power output and thermodynamic efficiency, my selected design will have a compression ratio of 19, optimizing for both the efficiency and net power output. This compression ratio yields the performance in Table 2.

Table 2 Efficiency & Net Power for $r_p = 19$, $\eta_c = 0.87$, $\eta_t = 0.85$

Thermodynamic Efficiency η_{th} (%)	Net Power Output (MW)
28.2	31.4

Final Answers. All the final answers required by this assignment are laid out below in Table 3.

Result	Value
Real Compressor Efficiency (η_c)	87%
Real Turbine Efficiency (η_t)	85%
Optimal Thermodynamic Efficiency (η_{th})	28.2%
Compression Ratio (r_p)	19
Mass Flow Rate ($\dot{m}$, $\frac{kg}{s}$)	249.3
Net Power Output (MW)	31.4
Heat Required Input (MW)	111.4
# of Heliostats	3510

Table 3 Final Answers

Sanity Check. The first number I will use for a sanity check is the number of heliostats. Coming in at 3510 heliostats, covering 351,000 square meters, this number is a bit alarming. However, according to the National Laboratory of the Rockies, "There can be more than 10,000 heliostats in a single CSP plant," which is conclusive enough for me that this makes sense [3]. Another sanity check I will perform is the mass flow rate. The mass flow rate calculation came out to be around 250kg/s, which seems a bit high. However, after researching some industrial power plant cycles produced by Siemens, they have at least 10 products with higher mass flow rates with the highest at over 1000kg/s. Therefore, I think it is reasonable to have a mass flow of 250kg/s [4]. Another thing to check is the total cycle efficiency, again I can use the same Siemens documentation. Siemens cites a simple power cycle gross efficiency ranging from $\sim$ 30% to $\sim$ 45%. Therefore, my efficiency is a little bit low when compared to these values, but I will argue that I am on the conservative side. The efficiencies I obtained may not be comparable to the efficiencies used in the Siemens products. This cycle passes all my sanity checks.

References

[1] Chapman, J. W., 2025, "Practical Techniques for Modeling Gas Turbine Engine Performance," NASA Glenn Research Center.

[2] Feb 2, 2026, "National Laboratories of the Rockies," https://www.nlr.gov/

[3] Feb 9, 2026, "National Laboratories of the Rockies," https://www.nlr.gov/grid/news/press/2021/nrel-launches-new-international-consortium-to-advance-high-tech-mirrors-used-in-solar-plants

[4] Feb 9, 2026, "Siemens Industrial Gas Turbines Specs," https://assets.new.siemens.com/siemens/assets/api/uuid:ab8578bf-d86f-45d9-a26b-7ac7a274fadd/siemens-gas-turbine-portfolio.pdf

Appendix: MATLAB Code

```
%written by _______ (id 528535) in conjunction with grok ai to assist with ...
    looping/indexing of the matrices

clear; clc; close all;

% Constants
cp    = 1.004e3;      % J/kg K
gamma = 1.4;
T1    = 300;          % K
T3    = 1200;         % K
T5    = 500;          % K
Q_out = 30e6;         % W  (30 MW heat out)

% compression ratio
r = linspace(5, 30, 250);

% The 7 efficiency combinations
efficiencies = [
    1.00  1.00
    1.00  0.90
    1.00  0.80
    0.90  1.00
    0.80  1.00
    0.90  0.90
    0.80  0.80
];

% Colors and line styles
colors = lines(7);
styles = {'-', '--', '-.', ':', '-', '--', '-.'};

% Preallocate arrays
r_opt_eta = zeros(size(efficiencies,1),1);
eta_max   = zeros(size(efficiencies,1),1);

% ...

% Figure 1: Thermal efficiency vs compression ratio
% ...

figure(1);
hold on; grid on; box on;

for i = 1:size(efficiencies,1)
    eta_t = efficiencies(i,1);
    eta_c = efficiencies(i,2);

    % Isentropic temperatures
    T2s = T1 .* r.^((gamma-1)/gamma);
    T4s = T3 .* (1./r).^((gamma-1)/gamma);

    % Real outlet temperatures
    T2 = T1 + (T2s - T1) / eta_c;
    T4 = T3 - eta_t .* (T3 - T4s);
```

```
52
53      % Specific quantities
54      q_in  = cp * (T3 - T2);
55      w_c   = cp * (T2 - T1);
56      w_t   = cp * (T3 - T4);
57      w_net = w_t - w_c;
58      eta_th = (w_net ./ q_in) * 100;     % percent
59
60      % Plot
61      plot(r, eta_th, 'Color', colors(i,:), ...
62           'LineStyle', styles{i}, 'LineWidth', 1.8, ...
63           'DisplayName', sprintf(' _t  = %.0f%%    _c  = %.0f%%', eta_t*100, ...
                eta_c*100));
64
65
66  end
67
68  xlabel('Compression ratio  r = p_2 / p_1');
69  ylabel('Thermal efficiency  \eta_{th}  (%)');
70  title('CCGT Topping Cycle     Thermal Efficiency vs Compression Ratio');
71  legend('Location','bestoutside','FontSize',10);
72  set(gca, 'FontSize',11);
73  ylim([-10 70]);
74  hold off;
75
76  % ...

77  % Figure 2: Net power output vs compression ratio
78  % ...

79  figure(2);
80  hold on; grid on; box on;
81
82  for i = 1:size(efficiencies,1)
83      eta_t = efficiencies(i,1);
84      eta_c = efficiencies(i,2);
85
86      % Isentropic temperatures
87      T2s = T1 .* r.^((gamma-1)/gamma);
88      T4s = T3 .* (1./r).^((gamma-1)/gamma);
89
90      % Real outlet temperatures
91      T2 = T1 + (T2s - T1) / eta_c;
92      T4 = T3 - eta_t .* (T3 - T4s);
93
94      % Specific quantities
95      q_out = cp * (T4 - T5);
96      w_net = cp * ((T3 - T4) - (T2 - T1));
97
98      m_dot = Q_out ./ q_out;           % kg/s
99      P_net = m_dot .* w_net / 1e6;     % MW   (note: /1e6 not /10e6)
100
101     % Plot
102     plot(r, P_net, 'Color', colors(i,:), ...
103          'LineStyle', styles{i}, 'LineWidth', 1.8, ...
104          'DisplayName', sprintf(' _t  = %.0f%%    _c  = %.0f%%', eta_t*100, ...
                eta_c*100));
```

```
end

xlabel('Compression ratio  r = p_2 / p_1');
ylabel('Net power output  P_{net}  (MW)');
title('CCGT Topping Cycle     Net Power Output vs Compression Ratio');
legend('Location','bestoutside','FontSize',10);
set(gca, 'FontSize',11);
ylim([-5e2,5e2])
hold off;



% ...

% SINGLE-CASE ANALYSIS SECTION
% ...


%one combination
eta_t_single = 0.85;      % turbine efficiency
eta_c_single = 0.87;      % compressor efficiency
% ================================================


%                  Calculations
T2s = T1 .* r.^((gamma-1)/gamma);
T4s = T3 .* (1./r).^((gamma-1)/gamma);

T2 = T1 + (T2s - T1) / eta_c_single;
T4 = T3 - eta_t_single .* (T3 - T4s);
temps = [T2;T4];
q_in  = cp * (T3 - T2);
q_out = cp * (T4 - T5);
w_net = cp * ((T3 - T4) - (T2 - T1));

eta_th = w_net ./ q_in * 100;           % percent

m_dot = Q_out ./ q_out;                 % kg/s
P_net = m_dot .* w_net / 1e6;           % MW

% Find optimum (max efficiency point)
[eta_max_single, idx] = max(eta_th);
r_opt_single = r(idx);
P_at_opt = P_net(idx);

%                  Figure 3: Single-case Thermal Efficiency
figure(3); clf; hold on; grid on; box on;

plot(r, eta_th, 'Color', [0 0.447 0.741], ...
     'LineWidth', 2.8, 'DisplayName', sprintf(' _t  = %.1f%%    _c  = %.1f%%', ...
     eta_t_single*100, eta_c_single*100));

xlabel('Compression ratio  r = p_2 / p_1');
ylabel('Thermal efficiency  \eta_{th}  (%)');
title(sprintf('Thermal Efficiency      _t  = %.1f%%   _c  = %.1f%%', ...
      eta_t_single*100, eta_c_single*100), 'FontSize',13);
```

```
160 legend('Location','best','FontSize',11);
161 xlim([min(r) max(r)]); ylim([0 55]);
162 set(gca, 'FontSize',12);
163 hold off;
164
165 %                 Figure 4: Single-case Net Power Output
166 figure(4); clf; hold on; grid on; box on;
167
168 plot(r, P_net, 'Color', [0.85 0.33 0.098], ...
169      'LineWidth', 2.8, 'DisplayName', sprintf(' _t  = %.1f%%    _c  = %.1f%%', ...
170      eta_t_single*100, eta_c_single*100));
171
172
173 xlabel('Compression ratio  r = p_2 / p_1');
174 ylabel('Net power output  P_{net}  (MW)');
175 title(sprintf('Net Power Output      _t  = %.1f%%   _c  = %.1f%%', ...
176       eta_t_single*100, eta_c_single*100), 'FontSize',13);
177 legend('Location','best','FontSize',11);
178 xlim([min(r) max(r)]);
179 set(gca, 'FontSize',12);
180 hold off;
```